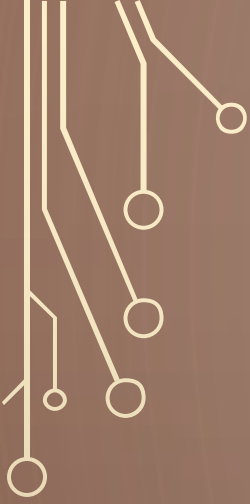




HABOTCONNECT HIRING PROJECT

SECURE CLOUD INFRASTRUCTURE PROVISIONING
USING TERRAFORM, GOOGLE CLOUD PLATFORM,
GITHUB ACTIONS & DJANGO REST FRAMEWORK

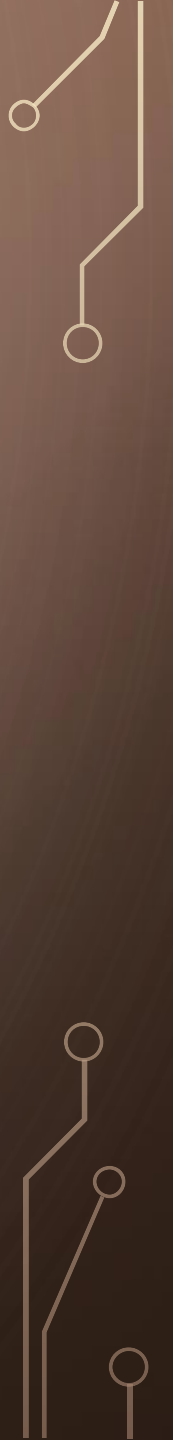
AMRUTA SURYAVANSHI



PROJECT OVERVIEW

- • Provision secure GCP staging infrastructure
- Automate deployment using Terraform
- Protect student data
- Implement Fail-Closed CI/CD

PROJECT OBJECTIVES

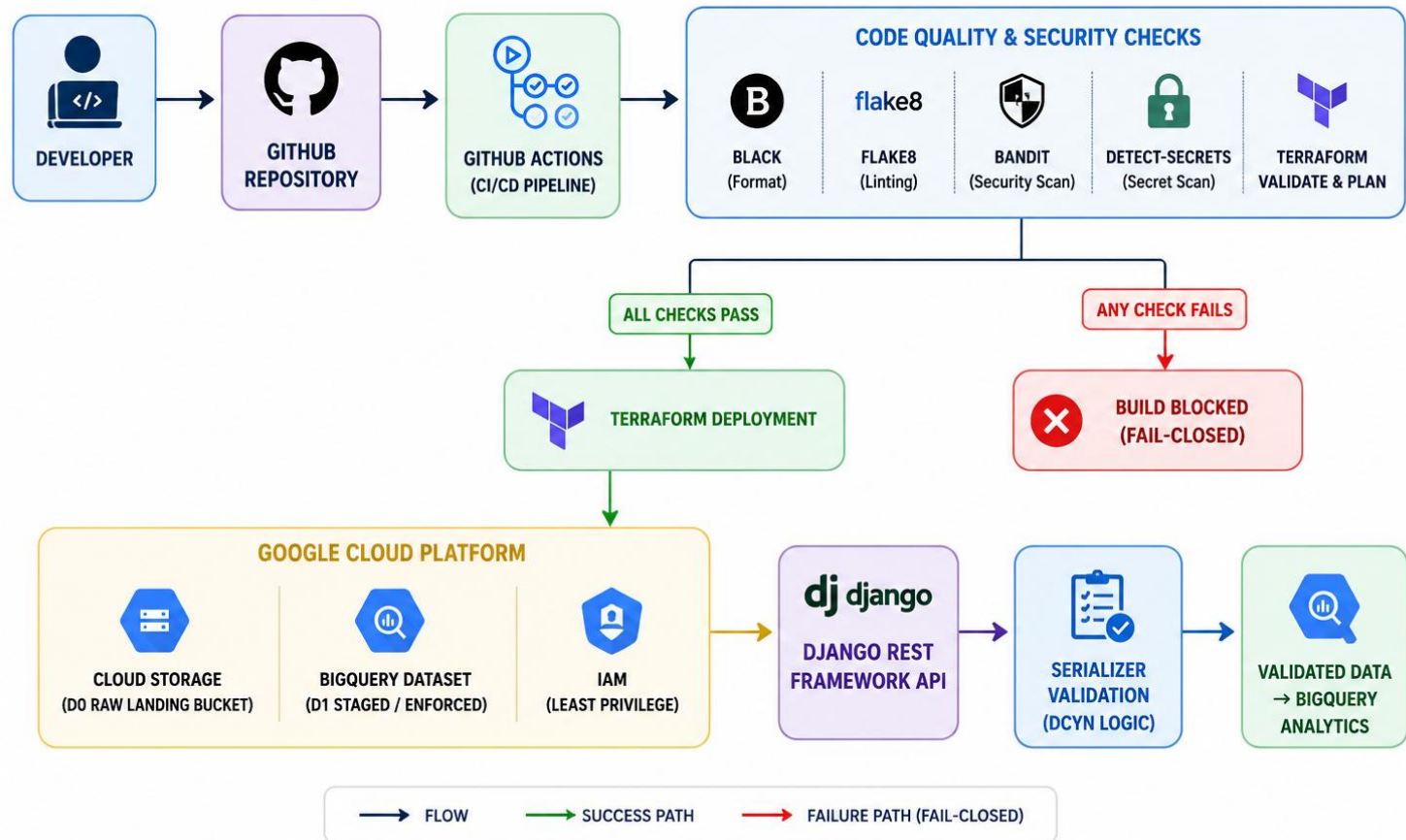
- Infrastructure as Code
 - Secure IAM & Least Privilege
 - Row-Level Security
 - Automated validation
- 
- 

TECHNOLOGY STACK

- Terraform
 - Google Cloud Platform
 - GitHub Actions
 - Django REST Framework
 - BigQuery
 - Cloud Storage

PROJECT ARCHITECTURE

PROJECT ARCHITECTURE





TASK 1: TERRAFORM

- • Provider & Variables
 - GCS Raw Landing Bucket
 - BigQuery Dataset
 - IAM
 - Public Access Prevention
 - Versioning
- 
- 
- 

SECURITY IMPLEMENTATION

- • IAM (RBAC)
 - Principle of Least Privilege
 - Row-Level Security (RLS)
 - Secret Protection

TASK 2: CI/CD

- Pipeline:

Checkout → terraform fmt → validate → flake8 →
Secret Scan → Deploy

Successful pipeline

The screenshot displays the GitHub Actions interface for a workflow named 'Habot Secure CI Pipeline'. The workflow is triggered by a push to the 'main' branch. The job 'security-checks' has completed successfully, with a total duration of 23 seconds. The workflow file is named 'ci.yml' and is triggered on push.

Summary of the workflow run:

Triggered via	Status	Total duration	Artifacts
SuryavanshiAmruta pushed -> 606b0f7 main	Success	23s	-

Run details:

Job	Status	Duration
security-checks	Success	14s

FAIL-CLOSED BUILD GATE

- Pipeline immediately stops if:
 - Formatting fails
 - Linting fails
 - Hardcoded API key detected
 - Terraform validation fails

TASK 3: DRF VALIDATION

- • Student onboarding JSON
 - Serializer validation
 - DCYN Yes/No logic
 - Strict field validation

TESTING & RESULTS

- Terraform Apply Successful
- Bucket & Dataset Created
- IAM Verified
- Pipeline Tested
- API Validation Successful

Terraform Apply

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + v [ ] [X] ... | [ ] [X]
```

```
google_bigquery_dataset.staging: Creation complete after 3s [id=projects/habot-devops-project-504219/datasets/d1_staged]  
google_storage_bucket.raw_landing: Creation complete after 4s [id=habot-raw-landing-504219]  
google_storage_bucket_iam_member.viewer: Creating...  
google_storage_bucket_iam_member.viewer: Creation complete after 6s [id=b/habot-raw-landing-504219/roles/storage.objectViewer/user:amrutasuryavanshi212@gmail.com]  
  
Apply complete! Resources: 3 added, 0 changed, 0 destroyed.  
  
Outputs:  
  
bucket_name = "habot-raw-landing-504219"  
dataset_name = "d1_staged"  
PS D:\Devops_projects\habot-devops-project\terraform> █
```

Storage bucket

Buckets

Create

Refresh

Go to path

Learn

Filter

Filter buckets

?

⚙

	Name ↑	Created	Location type	Location	Default storage class ?	Last modified
	habot-raw-landing-504219	Aug 2, 2026, 1:41:59 AM	Multi-region	asia	Standard	Aug 2, 2026, 1:41:59 AM

BigQuery dataset

IAM Verified

Permissions

[View by principals](#) [View by roles](#)

[+ Grant access](#) [- Remove access](#)

Filter Enter property name or value

Type	Principal ↑	Name	Role
☐	 amrutasuryavanshi212@gmail.com	Amruta Suryavanshi	Storage Object Viewer

Pipeline Tested

SuryavanshiAmruta / habot-devops-project

Code Issues Pull requests **Actions** Projects Wiki Security and quality Insights Settings

← Habot Secure CI Pipeline

✓ updated main.tf #39 [Re-run all jobs](#)

Summary

All jobs

✓ security-checks

Run details

Usage

Workflow file

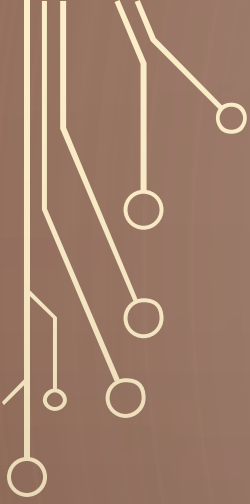
Triggered via push now

 SuryavanshiAmruta pushed to 606b0f7 main

Status **Success** Total duration **23s** Artifacts -

ci.yml on: push

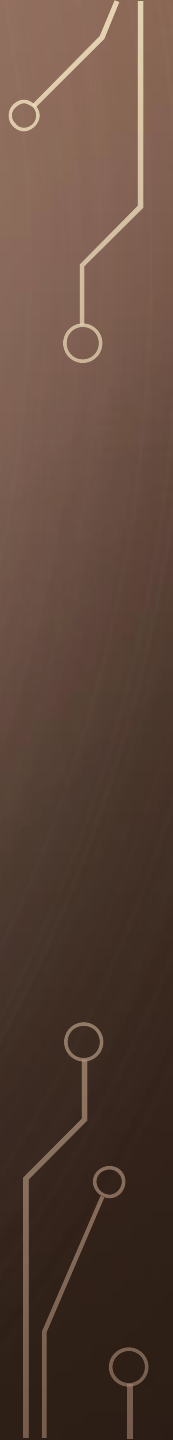
✓ security-checks 14s



CHALLENGES

- Learning IAM & RLS
- CI/CD integration
- Cloud security

FUTURE IMPROVEMENTS

- Remote Terraform state
 - Secret Manager
 - Cloud Monitoring
 - Multi-environment deployment
- 
- 

CONCLUSION

- Project meets HaborConnect requirements using secure IaC, automated CI/CD, robust validation, and cloud security best practices.

The background is a dark brown gradient with faint, concentric circles centered around the text. In the corners, there are stylized, light-colored lines resembling circuit traces or a network diagram, with small circles at the end of the lines.

THANK YOU